
Вариант 1.

$$f(x) = 0100000001001010$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 2.

$$f(x) = 0011101111110110$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 3.

$$f(x) = 0010011011111100$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 4.

$$f(x) = 1111010110110111$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 5.

$$f(x) = 0010000101011101$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 6.

$$f(x) = 1101011001001100$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 7.

$$f(x) = 1000100111010011$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 8.

$$f(x) = 0110010010100000$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 9.

$$f(x) = 0111101110001111$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 10.

$$f(x) = 0011001101000010$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 11.

$$f(x) = 0110110011000000$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Вариант 12.

$$f(x) = 0011010100000100$$

1. Построить АНФ преобразованием Мёбиуса.
 2. Найти нелинейность N_f .
 3. Найти все линейные статистические аналоги.
-

Ответы

1 0100000001001010

АНФ: $x_1x_2x_3 + x_1x_3x_4 + x_1x_3 + x_1x_4 + x_1 + x_2x_3$

$N_f = 4$

Аналоги: $\langle 0000, x \rangle, \langle 0101, x \rangle$

u	$W_f(u)$	
0000	8	$W_f(0000) = 12 - 4 = 8$
0001	0	$W_f(0001) = 8 - 8 = 0$
0010	4	$W_f(0010) = 6 - 10 = 4$
0011	4	$W_f(0011) = 10 - 6 = 4$
0100	0	$W_f(0100) = 8 - 8 = 0$
0101	0	$W_f(0101) = 12 - 4 = 8$
0110	0	$W_f(0110) = 6 - 10 = -4$
0111	8	$W_f(0111) = 10 - 6 = -4$
1000	-4	$W_f(1000) = 10 - 6 = -4$
1001	-4	$W_f(1001) = 10 - 6 = 4$
1010	-4	$W_f(1010) = 8 - 8 = 0$
1011	-4	$W_f(1011) = 8 - 8 = 0$
1100	4	$W_f(1100) = 6 - 10 = -4$
1101	0	$W_f(1101) = 6 - 10 = 4$
1110	0	$W_f(1110) = 8 - 8 = 0$
1111	0	$W_f(1111) = 8 - 8 = 0$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	1
0010	0
0011	1
0100	0
0101	1
0110	0
0111	1
1000	0
1001	0
1010	0
1011	0
1100	1
1101	1
1110	0
1111	0

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 1$
 $c_{0010} = f(0000) \oplus f(0010) = 0$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 1$
 $c_{0100} = f(0000) \oplus f(0100) = 0$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 1$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 0$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 1$
 $c_{1000} = f(0000) \oplus f(1000) = 0$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 0$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 0$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 1$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 0$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 0$

2 0011101111110110

АНФ: $x_1x_2x_3x_4 + x_1x_2x_4 + x_1x_3x_4 + x_1x_3 + x_1x_4 + x_1 + x_2x_3x_4 + x_2x_3 + x_2 + x_3x_4 + x_3 + x_4$

$N_f = 5$

Аналоги: $1 + \langle 0000, x \rangle, 1 + \langle 0010, x \rangle, \langle 1010, x \rangle, \langle 1011, x \rangle, \langle 1100, x \rangle, \langle 1111, x \rangle$

u	$W_f(u)$	
0000	-6	$W_f(0000) = 5 - 11 = -6$
0001	2	$W_f(0001) = 7 - 9 = 2$
0010	-6	$W_f(0010) = 11 - 5 = -6$
0011	2	$W_f(0011) = 9 - 7 = 2$
0100	2	$W_f(0100) = 7 - 9 = 2$
0101	2	$W_f(0101) = 9 - 7 = 2$
0110	2	$W_f(0110) = 9 - 7 = 2$
0111	2	$W_f(0111) = 7 - 9 = 2$
1000	-2	$W_f(1000) = 9 - 7 = -2$
1001	-2	$W_f(1001) = 7 - 9 = -2$
1010	6	$W_f(1010) = 11 - 5 = 6$
1011	6	$W_f(1011) = 5 - 11 = 6$
1100	6	$W_f(1100) = 11 - 5 = 6$
1101	-2	$W_f(1101) = 9 - 7 = -2$
1110	-2	$W_f(1110) = 9 - 7 = -2$
1111	6	$W_f(1111) = 11 - 5 = 6$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	0
0010	1
0011	0
0100	1
0101	1
0110	1
0111	1
1000	1
1001	0
1010	1
1011	0
1100	0
1101	0
1110	0
1111	1

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 0$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 0$
 $c_{0100} = f(0000) \oplus f(0100) = 1$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 1$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 1$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 1$
 $c_{1000} = f(0000) \oplus f(1000) = 1$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 0$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 1$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 0$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 0$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 0$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 1$

3 0010011011111100

АНФ: $x_1x_2x_3x_4 + x_1x_4 + x_2x_3 + x_2 + x_3x_4 + x_3$

$N_f = 5$

Аналоги: $\langle 0011, x \rangle$, $\langle 0110, x \rangle$, $1 + \langle 1000, x \rangle$, $\langle 1010, x \rangle$, $1 + \langle 1011, x \rangle$, $\langle 1100, x \rangle$

u	$W_f(u)$	
0000	-2	$W_f(0000) = 7 - 9 = -2$
0001	2	$W_f(0001) = 7 - 9 = 2$
0010	2	$W_f(0010) = 7 - 9 = 2$
0011	6	$W_f(0011) = 11 - 5 = 6$
0100	2	$W_f(0100) = 7 - 9 = 2$
0101	6	$W_f(0101) = 7 - 9 = -2$
0110	2	$W_f(0110) = 11 - 5 = 6$
0111	-2	$W_f(0111) = 7 - 9 = 2$
1000	6	$W_f(1000) = 11 - 5 = -6$
1001	-2	$W_f(1001) = 7 - 9 = -2$
1010	6	$W_f(1010) = 11 - 5 = 6$
1011	-6	$W_f(1011) = 11 - 5 = -6$
1100	-2	$W_f(1100) = 11 - 5 = 6$
1101	6	$W_f(1101) = 7 - 9 = 2$
1110	-6	$W_f(1110) = 7 - 9 = 2$
1111	6	$W_f(1111) = 7 - 9 = -2$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	0
0010	1
0011	1
0100	0
0101	1
0110	0
0111	1
1000	1
1001	0
1010	1
1011	1
1100	0
1101	1
1110	1
1111	1

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 0$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 1$
 $c_{0100} = f(0000) \oplus f(0100) = 0$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 1$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 0$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 1$
 $c_{1000} = f(0000) \oplus f(1000) = 1$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 0$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 1$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 1$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 0$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 1$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 1$

4 1111010110110111

АНФ: $x_1x_2x_4 + x_1x_2 + x_1x_3x_4 + x_1x_4 + x_2x_3x_4 + x_2x_3 + 1$

$N_f = 4$

Аналоги: $1 + \langle 0000, x \rangle, 1 + \langle 0101, x \rangle$

u	$W_f(u)$	
0000	-8	$W_f(0000) = 4 - 12 = -8$
0001	-4	$W_f(0001) = 10 - 6 = -4$
0010	-4	$W_f(0010) = 10 - 6 = -4$
0011	0	$W_f(0011) = 8 - 8 = 0$
0100	4	$W_f(0100) = 6 - 10 = 4$
0101	-8	$W_f(0101) = 4 - 12 = -8$
0110	0	$W_f(0110) = 8 - 8 = 0$
0111	4	$W_f(0111) = 6 - 10 = 4$
1000	0	$W_f(1000) = 8 - 8 = 0$
1001	4	$W_f(1001) = 10 - 6 = 4$
1010	-4	$W_f(1010) = 6 - 10 = -4$
1011	0	$W_f(1011) = 8 - 8 = 0$
1100	-4	$W_f(1100) = 6 - 10 = -4$
1101	0	$W_f(1101) = 8 - 8 = 0$
1110	0	$W_f(1110) = 8 - 8 = 0$
1111	4	$W_f(1111) = 10 - 6 = 4$

Преобр. Мёбиуса	
g	c_g
0000	1
0001	0
0010	0
0011	0
0100	1
0101	1
0110	0
0111	0
1000	0
1001	1
1010	0
1011	1
1100	0
1101	1
1110	1
1111	0

$c_{0000} = f(0000) = 1$
 $c_{0001} = f(0000) \oplus f(0001) = 0$
 $c_{0010} = f(0000) \oplus f(0010) = 0$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 0$
 $c_{0100} = f(0000) \oplus f(0100) = 1$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 1$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 0$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 0$
 $c_{1000} = f(0000) \oplus f(1000) = 0$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 1$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 0$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 1$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 0$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 1$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 0$

5 0010000101011101

АНФ: $x_1x_2x_3x_4 + x_1x_2x_3 + x_1x_2 + x_1 + x_2x_3x_4 + x_2x_4 + x_3x_4 + 1$

$N_f = 5$

Аналоги: $1 + \langle 0001, x \rangle$, $1 + \langle 0111, x \rangle$, $1 + \langle 1000, x \rangle$, $1 + \langle 1001, x \rangle$, $\langle 1010, x \rangle$, $\langle 1101, x \rangle$

u	$W_f(u)$	
0000	2	$W_f(0000) = 9 - 7 = 2$
0001	-6	$W_f(0001) = 11 - 5 = -6$
0010	-2	$W_f(0010) = 9 - 7 = -2$
0011	-2	$W_f(0011) = 7 - 9 = -2$
0100	-2	$W_f(0100) = 9 - 7 = -2$
0101	-2	$W_f(0101) = 7 - 9 = -2$
0110	-2	$W_f(0110) = 9 - 7 = 2$
0111	-2	$W_f(0111) = 11 - 5 = -6$
1000	2	$W_f(1000) = 11 - 5 = -6$
1001	-6	$W_f(1001) = 5 - 11 = -6$
1010	-6	$W_f(1010) = 11 - 5 = 6$
1011	-6	$W_f(1011) = 9 - 7 = -2$
1100	-6	$W_f(1100) = 7 - 9 = -2$
1101	6	$W_f(1101) = 5 - 11 = 6$
1110	-2	$W_f(1110) = 7 - 9 = 2$
1111	-2	$W_f(1111) = 9 - 7 = 2$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	0
0010	1
0011	1
0100	0
0101	0
0110	1
0111	0
1000	0
1001	1
1010	1
1011	1
1100	1
1101	1
1110	0
1111	1

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 0$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 1$
 $c_{0100} = f(0000) \oplus f(0100) = 0$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 0$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 1$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 0$
 $c_{1000} = f(0000) \oplus f(1000) = 0$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 1$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 1$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 1$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 1$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 0$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 1$

6 1101011001001100

АНФ: $x_1x_2x_3 + x_1x_4 + x_2x_3x_4 + x_2 + x_3x_4$

$N_f = 4$

Аналоги: $\langle 0010, x \rangle, 1 + \langle 1111, x \rangle$

u	$W_f(u)$	
0000	0	$W_f(0000) = 8 - 8 = 0$
0001	-4	$W_f(0001) = 10 - 6 = -4$
0010	8	$W_f(0010) = 4 - 12 = 8$
0011	4	$W_f(0011) = 10 - 6 = 4$
0100	0	$W_f(0100) = 8 - 8 = 0$
0101	4	$W_f(0101) = 10 - 6 = 4$
0110	0	$W_f(0110) = 8 - 8 = 0$
0111	4	$W_f(0111) = 6 - 10 = 4$
1000	4	$W_f(1000) = 6 - 10 = 4$
1001	0	$W_f(1001) = 8 - 8 = 0$
1010	4	$W_f(1010) = 10 - 6 = 4$
1011	0	$W_f(1011) = 8 - 8 = 0$
1100	-4	$W_f(1100) = 6 - 10 = -4$
1101	0	$W_f(1101) = 8 - 8 = 0$
1110	4	$W_f(1110) = 6 - 10 = 4$
1111	-8	$W_f(1111) = 4 - 12 = -8$

Преобр. Мёбиуса	
g	c_g
0000	1
0001	0
0010	1
0011	1
0100	1
0101	1
0110	0
0111	1
1000	1
1001	1
1010	1
1011	0
1100	0
1101	0
1110	1
1111	0

$c_{0000} = f(0000) = 1$
 $c_{0001} = f(0000) \oplus f(0001) = 0$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 1$
 $c_{0100} = f(0000) \oplus f(0100) = 1$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 1$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 0$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 1$
 $c_{1000} = f(0000) \oplus f(1000) = 1$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 1$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 1$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 0$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 0$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 1$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 0$

7 1000100111010011

АНФ: $x_1x_2x_3 + x_1x_3 + x_1x_4 + x_2x_3 + x_2 + x_3x_4 + 1$

$N_f = 4$

Аналоги: $1 + \langle 0011, x \rangle, 1 + \langle 0110, x \rangle$

u	$W_f(u)$	
0000	0	$W_f(0000) = 8 - 8 = 0$
0001	0	$W_f(0001) = 8 - 8 = 0$
0010	0	$W_f(0010) = 8 - 8 = 0$
0011	-8	$W_f(0011) = 4 - 12 = -8$
0100	0	$W_f(0100) = 8 - 8 = 0$
0101	0	$W_f(0101) = 8 - 8 = 0$
0110	0	$W_f(0110) = 4 - 12 = -8$
0111	0	$W_f(0111) = 8 - 8 = 0$
1000	-8	$W_f(1000) = 10 - 6 = -4$
1001	0	$W_f(1001) = 6 - 10 = -4$
1010	-4	$W_f(1010) = 6 - 10 = -4$
1011	0	$W_f(1011) = 6 - 10 = 4$
1100	-4	$W_f(1100) = 6 - 10 = 4$
1101	-4	$W_f(1101) = 10 - 6 = 4$
1110	4	$W_f(1110) = 10 - 6 = -4$
1111	4	$W_f(1111) = 10 - 6 = 4$

Преобр. Мёбиуса	
g	c_g
0000	1
0001	1
0010	1
0011	1
0100	0
0101	0
0110	0
0111	1
1000	0
1001	1
1010	0
1011	0
1100	1
1101	0
1110	0
1111	0

$c_{0000} = f(0000) = 1$
 $c_{0001} = f(0000) \oplus f(0001) = 1$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 1$
 $c_{0100} = f(0000) \oplus f(0100) = 0$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 0$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 0$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 1$
 $c_{1000} = f(0000) \oplus f(1000) = 0$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 1$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 0$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 1$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 0$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 0$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 0$

8 0110010010100000

АНФ: $x_1x_2x_3x_4 + x_1x_2x_4 + x_1x_3 + x_2x_4$

$N_f = 5$

Аналоги: $\langle 0000, x \rangle, \langle 0011, x \rangle, \langle 0100, x \rangle, 1 + \langle 0101, x \rangle, \langle 1001, x \rangle, 1 + \langle 1011, x \rangle$

u	$W_f(u)$	
0000	6	$W_f(0000) = 11 - 5 = 6$
0001	2	$W_f(0001) = 7 - 9 = 2$
0010	2	$W_f(0010) = 7 - 9 = 2$
0011	6	$W_f(0011) = 11 - 5 = 6$
0100	6	$W_f(0100) = 5 - 11 = 6$
0101	-6	$W_f(0101) = 5 - 11 = -6$
0110	6	$W_f(0110) = 9 - 7 = 2$
0111	-6	$W_f(0111) = 9 - 7 = -2$
1000	2	$W_f(1000) = 7 - 9 = 2$
1001	6	$W_f(1001) = 11 - 5 = 6$
1010	-2	$W_f(1010) = 7 - 9 = -2$
1011	-6	$W_f(1011) = 11 - 5 = -6$
1100	2	$W_f(1100) = 9 - 7 = 2$
1101	-2	$W_f(1101) = 9 - 7 = -2$
1110	-6	$W_f(1110) = 9 - 7 = -2$
1111	2	$W_f(1111) = 9 - 7 = 2$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	1
0010	1
0011	0
0100	0
0101	0
0110	1
0111	1
1000	1
1001	0
1010	1
1011	0
1100	1
1101	1
1110	1
1111	1

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 1$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 0$
 $c_{0100} = f(0000) \oplus f(0100) = 0$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 0$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 1$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 1$
 $c_{1000} = f(0000) \oplus f(1000) = 1$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 0$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 1$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 1$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 1$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 1$

9 0111101110001111

АНФ: $x_1x_2x_3x_4 + x_1x_2x_3 + x_1x_2x_4 + x_2x_3x_4 + x_2x_4 + x_3x_4 + x_3 + 1$

$N_f = 5$

Аналоги: $1 + \langle 0000, x \rangle$, $1 + \langle 0100, x \rangle$, $\langle 1010, x \rangle$, $1 + \langle 1100, x \rangle$, $1 + \langle 1101, x \rangle$, $\langle 1111, x \rangle$

u	$W_f(u)$	
0000	-6	$W_f(0000) = 5 - 11 = -6$
0001	2	$W_f(0001) = 7 - 9 = 2$
0010	-2	$W_f(0010) = 9 - 7 = -2$
0011	-2	$W_f(0011) = 7 - 9 = -2$
0100	-6	$W_f(0100) = 11 - 5 = -6$
0101	-2	$W_f(0101) = 9 - 7 = 2$
0110	-6	$W_f(0110) = 7 - 9 = -2$
0111	2	$W_f(0111) = 9 - 7 = -2$
1000	-2	$W_f(1000) = 7 - 9 = 2$
1001	-2	$W_f(1001) = 9 - 7 = 2$
1010	2	$W_f(1010) = 11 - 5 = 6$
1011	2	$W_f(1011) = 9 - 7 = -2$
1100	2	$W_f(1100) = 5 - 11 = -6$
1101	6	$W_f(1101) = 11 - 5 = -6$
1110	-2	$W_f(1110) = 9 - 7 = -2$
1111	-6	$W_f(1111) = 11 - 5 = 6$
1101	-6	
1110	-2	
1111	6	

Преобр. Мёбиуса	
g	c_g
0000	0
0001	1
0010	1
0011	1
0100	1
0101	0
0110	1
0111	0
1000	1
1001	0
1010	0
1011	0
1100	1
1101	1
1110	0
1111	1

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 1$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 1$
 $c_{0100} = f(0000) \oplus f(0100) = 1$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 0$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 1$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 0$
 $c_{1000} = f(0000) \oplus f(1000) = 1$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 0$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 0$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 1$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 0$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 1$

10 0011001101000010

АНФ: $x_1x_2x_4 + x_1x_2 + x_1x_3x_4 + x_1x_3 + x_1x_4 + x_1 + x_2x_3x_4 + x_2x_3 + x_2x_4 + x_4$

$N_f = 4$

Аналоги: $1 + \langle 0010, x \rangle, \langle 1010, x \rangle$

u	$W_f(u)$	
0000	4	$W_f(0000) = 10 - 6 = 4$
0001	0	$W_f(0001) = 8 - 8 = 0$
0010	-8	$W_f(0010) = 12 - 4 = -8$
0011	4	$W_f(0011) = 10 - 6 = 4$
0100	4	$W_f(0100) = 8 - 8 = 0$
0101	0	$W_f(0101) = 10 - 6 = 4$
0110	0	$W_f(0110) = 6 - 10 = -4$
0111	4	$W_f(0111) = 8 - 8 = 0$
1000	-4	$W_f(1000) = 6 - 10 = 4$
1001	0	$W_f(1001) = 8 - 8 = 0$
1010	4	$W_f(1010) = 12 - 4 = 8$
1011	0	$W_f(1011) = 6 - 10 = 4$
1100	0	$W_f(1100) = 8 - 8 = 0$
1101	8	$W_f(1101) = 6 - 10 = 4$
1110	4	$W_f(1110) = 10 - 6 = -4$
1111	0	$W_f(1111) = 8 - 8 = 0$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	0
0010	1
0011	0
0100	0
0101	0
0110	0
0111	0
1000	0
1001	1
1010	1
1011	1
1100	0
1101	1
1110	1
1111	0

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 0$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 0$
 $c_{0100} = f(0000) \oplus f(0100) = 0$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 0$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 0$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 0$
 $c_{1000} = f(0000) \oplus f(1000) = 0$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 1$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 1$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 1$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 0$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 1$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 0$

11 0110110011000000

АНФ: $x_1x_3x_4 + x_2x_3x_4 + x_2x_3 + x_2x_4$

$N_f = 4$

Аналоги: $\langle 0010, x \rangle, 1 + \langle 1110, x \rangle$

u	$W_f(u)$	
0000	4	$W_f(0000) = 10 - 6 = 4$
0001	0	$W_f(0001) = 8 - 8 = 0$
0010	8	$W_f(0010) = 4 - 12 = 8$
0011	4	$W_f(0011) = 10 - 6 = 4$
0100	4	$W_f(0100) = 6 - 10 = 4$
0101	4	$W_f(0101) = 8 - 8 = 0$
0110	4	$W_f(0110) = 8 - 8 = 0$
0111	0	$W_f(0111) = 10 - 6 = -4$
1000	0	$W_f(1000) = 6 - 10 = 4$
1001	0	$W_f(1001) = 8 - 8 = 0$
1010	0	$W_f(1010) = 8 - 8 = 0$
1011	-4	$W_f(1011) = 10 - 6 = -4$
1100	4	$W_f(1100) = 10 - 6 = 4$
1101	0	$W_f(1101) = 8 - 8 = 0$
1110	-8	$W_f(1110) = 12 - 4 = -8$
1111	4	$W_f(1111) = 10 - 6 = 4$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	1
0010	1
0011	0
0100	1
0101	1
0110	0
0111	0
1000	1
1001	1
1010	0
1011	0
1100	0
1101	1
1110	1
1111	0

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 1$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 0$
 $c_{0100} = f(0000) \oplus f(0100) = 1$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 1$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 0$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 0$
 $c_{1000} = f(0000) \oplus f(1000) = 1$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 1$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 0$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 0$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 1$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 1$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 0$

12 0011010100000100

АНФ: $x_1x_2x_3x_4 + x_1x_2x_3 + x_1x_2x_4 + x_1x_2 + x_1x_3x_4 + x_1x_4 + x_2x_3 + x_2x_4 + x_2 + x_4$

$N_f = 5$

Аналоги: $\langle 0000, x \rangle$, $1 + \langle 0001, x \rangle$, $1 + \langle 0101, x \rangle$, $\langle 0110, x \rangle$, $\langle 1000, x \rangle$, $\langle 1010, x \rangle$

u	$W_f(u)$	
0000	6	$W_f(0000) = 11 - 5 = 6$
0001	-6	$W_f(0001) = 11 - 5 = -6$
0010	-2	$W_f(0010) = 9 - 7 = -2$
0011	2	$W_f(0011) = 9 - 7 = 2$
0100	-2	$W_f(0100) = 9 - 7 = -2$
0101	2	$W_f(0101) = 5 - 11 = -6$
0110	-2	$W_f(0110) = 11 - 5 = 6$
0111	-6	$W_f(0111) = 7 - 9 = 2$
1000	6	$W_f(1000) = 5 - 11 = 6$
1001	2	$W_f(1001) = 9 - 7 = 2$
1010	6	$W_f(1010) = 11 - 5 = 6$
1011	2	$W_f(1011) = 7 - 9 = 2$
1100	-2	$W_f(1100) = 7 - 9 = -2$
1101	2	$W_f(1101) = 7 - 9 = 2$
1110	-2	$W_f(1110) = 9 - 7 = -2$
1111	2	$W_f(1111) = 9 - 7 = 2$

Преобр. Мёбиуса	
g	c_g
0000	0
0001	0
0010	1
0011	0
0100	0
0101	1
0110	1
0111	0
1000	0
1001	0
1010	1
1011	0
1100	0
1101	0
1110	1
1111	1

$c_{0000} = f(0000) = 0$
 $c_{0001} = f(0000) \oplus f(0001) = 0$
 $c_{0010} = f(0000) \oplus f(0010) = 1$
 $c_{0011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) = 0$
 $c_{0100} = f(0000) \oplus f(0100) = 0$
 $c_{0101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) = 1$
 $c_{0110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) = 1$
 $c_{0111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) = 0$
 $c_{1000} = f(0000) \oplus f(1000) = 0$
 $c_{1001} = f(0000) \oplus f(0001) \oplus f(1000) \oplus f(1001) = 0$
 $c_{1010} = f(0000) \oplus f(0010) \oplus f(1000) \oplus f(1010) = 1$
 $c_{1011} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) = 0$
 $c_{1100} = f(0000) \oplus f(0100) \oplus f(1000) \oplus f(1100) = 0$
 $c_{1101} = f(0000) \oplus f(0001) \oplus f(0100) \oplus f(0101) \oplus f(1000) \oplus f(1001) \oplus f(1100) \oplus f(1101) = 0$
 $c_{1110} = f(0000) \oplus f(0010) \oplus f(0100) \oplus f(0110) \oplus f(1000) \oplus f(1010) \oplus f(1100) \oplus f(1110) = 1$
 $c_{1111} = f(0000) \oplus f(0001) \oplus f(0010) \oplus f(0011) \oplus f(0100) \oplus f(0101) \oplus f(0110) \oplus f(0111) \oplus f(1000) \oplus f(1001) \oplus f(1010) \oplus f(1011) \oplus f(1100) \oplus f(1101) \oplus f(1110) \oplus f(1111) = 1$